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Enhancing precision in two-dimensional lithium battery modeling using an improved discharge boundary condition setting within the lattice Boltzmann method framework for broad applications

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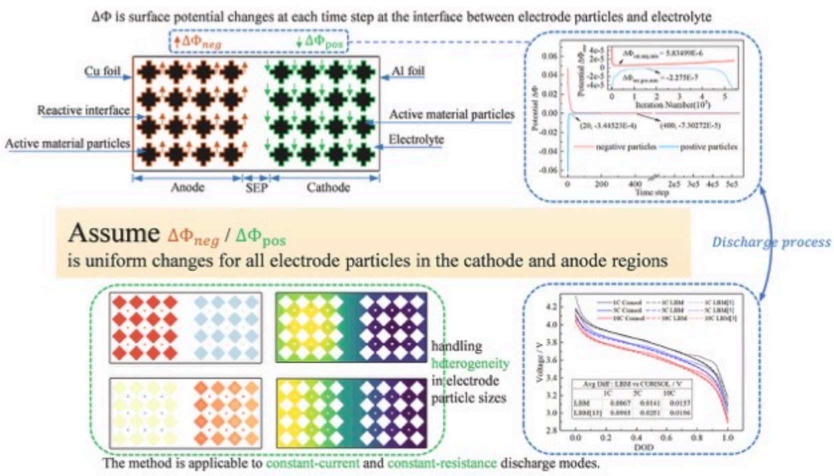
HIGHLIGHTS

- Proposed a new discharge boundary condition for 2D lithium battery models.
- Enhanced modeling accuracy within an optimized lattice Boltzmann method framework.
- Achieved current, charge, and mass conservation with uniform electrode potential changes.
- Model applicable to constant-resistance discharge and particle size heterogeneity.

GRAPHICAL ABSTRACT



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ABSTRACT

To enhance the modeling accuracy of the two-dimensional lithium battery model and improve its precision in predicting the discharge process, this study proposes an approach based on a new discharge boundary condition setting. In this method, when a battery is connected to a load circuit, electrochemical reactions occurring at the interface between the electrode particles and the electrolyte induce uniform changes in the surface potentials of all electrode particles in the cathode and anode regions, rather than being confined to the battery terminals near the current collectors. These changes are quantified based on the load current using an optimization algorithm, instead of conductivity calculations. Leveraging this approach, a two-dimensional battery model is developed within an improved lattice Boltzmann method framework. Simulation results reveal that the model satisfies current and charge conservation principles, substantially improves computational accuracy, and demonstrates high stability. Furthermore, the two-dimensional local variations in lithium/lithium-ion concentrations predicted

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by the model align well with the general mechanism analysis. Notably, this model demonstrates versatility for broad applications, effectively predicting battery behavior under varied discharge conditions, such as constant-current and constant-resistance discharge modes, and can handle electrode particle size heterogeneity.

1. Introduction

Lithium-ion batteries find widespread applications in consumer electronics, electric vehicles, and energy storage systems owing to their numerous advantages, including high energy densities, high charging efficiencies, and lack of memory effects [1]. However, exposure to frequent charging–discharging cycles or overcharging can shorten the cycle life of such batteries and compromise their reliability [2,3]. In extreme cases, such as thermal runaway, the batteries may even catch fire or explode [4].

The charging and discharging processes of lithium-ion batteries depend on the reversible migration and intercalation of lithium ions between the cathode and anode. These microscopic transport mechanisms are influenced by material properties and electrode morphology, which are intimately linked to the macroscopic performance of the battery [5–7]. However, direct observations or quantitative real-time analyses of these transport processes using current experimental techniques are challenging. Conversely, numerical simulations can offer insights into the internal charging–discharging processes of batteries, thereby contributing toward enhancements in their performance and reliability [8,9].

To comprehensively examine the influence of electrode particle size, shape, and distribution, as well as material properties, on the charging and discharging processes of lithium-ion batteries, Jiang et al. developed a two-dimensional lattice Boltzmann method (2D LBM) model for constant current discharge based on the pseudo-two-dimensional (P2D) theory [10,11] and the LBM approach for electrochemical systems [12], which offers excellent numerical stability and constitutive versatility [13,14]. However, the accuracy of this model was observed to be limited, particularly during the initial and final stages of discharge. This inadequate accuracy was attributed to the assumption of inappropriate boundary conditions and an oversimplified treatment of current conservation during model construction.

First, Jiang et al. assumed that the potential change induced by the discharge current propagates through the internal electrolyte to initiate the discharge process. However, electrolyte cannot conduct electrons directly [15]. Consequently, before electrochemical reactions begin, the potential within the electrolyte remains largely constant. In most battery simulations, boundary conditions are typically imposed on the surfaces of electrode particles intersecting with the terminals of the battery model [16,17].

Second, Jiang et al. failed to verify the relationship between the local reaction current density and load current density. Notably, discharge currents originate from electrons released by electrochemical reactions occurring on the surfaces of electrode particles. Consequently, iterative calculations must ensure that the sum of local reaction current densities in the anode region equals the discharge current density to guarantee current conservation [18].

Third, in another study [19], it was highlighted that Jiang et al. assumed a constant charge transfer resistance at the solid/electrolyte interface, representing a significant simplification that may not sufficiently capture the true interfacial behavior. To streamline the integration of the exponential Butler–Volmer expression in the LBM iteration process, they proposed treating electrode particles within a porous electrode as equipotential. However, this assumption became problematic under variations in electrode particle positions, sizes, or shapes, which are conditions typical of aging batteries [20]. Notable discrepancies are easily observed in the potentials at different surface locations on the heterogeneous electrode particles [14].

Notably, the constant current discharge process of a battery can be maintained by adjusting the external circuit load. To assess the impact of load variation on battery performance, current density must be calculated. However, in constant resistance discharge, where the discharge current is an unknown variable, the method proposed by Ref. [19] proves insufficient.

This study proposes a new method for setting discharge boundary conditions in the 2D battery model. The proposed method assumes that electrochemical reactions occurring at the electrode particle–electrolyte interface result in the same surface potential change across all electrode particles in the anode and, separately, in the cathode. An optimization algorithm is employed to ensure current conservation during iterative calculations. Using this approach, a modified 2D LBM framework is developed to evaluate the accuracy of the discharge voltage response. Subsequently, the effect of the external circuit load on the battery is analyzed by modeling it as a change in the surface potential of electrode particles, offering qualitative insights into the behavior of the battery under constant resistance discharge conditions. The study further explores the impact of electrode particle size heterogeneity on lithium/lithium-ion transport, potential, and reaction current density variations during the battery discharge process. The above analyses are conducted to demonstrate the broad application potential of the proposed method in addressing practical battery-related problems.

2. Two-dimensional battery model

Fig. 1 illustrates a schematic of a rechargeable lithium-ion battery, comprising three components: the cathode region, separator, and anode region. In this model, the cathode and anode particles are embedded in the electrolyte of their respective regions, and conductive particles connect the electrode particles.

The model makes the following assumptions:

1) Each electrode particle is connected through conductive particles, and some electrode particles are directly connected to the current collector via conductive agents.
2) The resistance of the conductive agent is extremely low and can be considered negligible.
3) All electrons generated by electrochemical reactions occurring on the surfaces of electrode particles can reach the current collector through the conductive agent.

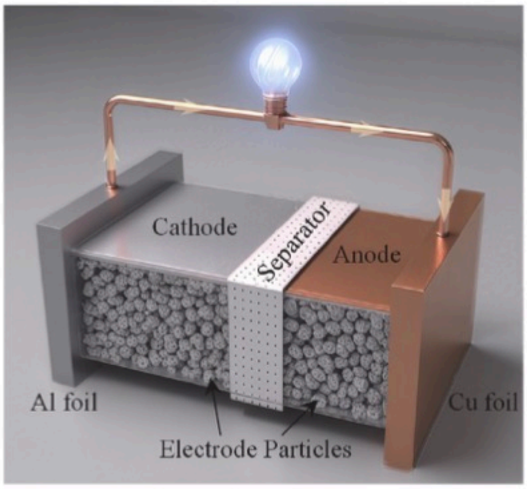

Fig. 1. Schematic of a rechargeable lithium-ion battery.

4) Electrode particles in both the cathode and anode regions exhibit identical electrical conductivity.
5) For simplicity, the resistance of the current collector is neglected.

Under these assumptions, the surfaces of the cathode and anode particles in their respective regions can be treated as two parallel circuits connected by conductive particles. When the battery is connected to an external load circuit, electrochemical reactions occur at the electrode particle–electrolyte interfaces in the cathode region. During these reactions, cathode particles release electrons, increasing the surface potential of all cathode particles by $\Delta\Phi_{neg}$. Concurrently, the released electrons traverse the external circuit and reach the electrode−electrolyte interface in the anode region, where they participate in further electrochemical reactions and combine with lithium ions to form lithium compounds. This causes the surface potential of all anode particles to drop by $\Delta\Phi_{pos}$. Simultaneously, the electrochemical reactions occurring at the cathode particle interface release lithium ions into the electrolyte. These ions migrate from the cathode region to the anode region via two pathways: diffusion and migration. The complete electrochemical reaction process at the interface can be described as follows, where $LiMn_2O_4$ denotes lithium manganese oxide as cathode active material, LiC_6 represents lithiated graphite as anode active material, Li^+ and e^- indicate lithium ions and electrons, respectively, DCH refers to the discharge process, and x is the stoichiometric number of lithium ions participating during discharge.:

$$Li_{(y-x)}Mn_2O_4 + xLi^+ + xe^- \xrightarrow{DCH} Li_yMn_2O_4, \tag{1}$$

$$Li_xC_6 \xrightarrow{DCH} C_6 + xLi^+ + xe^-. \tag{2}$$

The electrode particle potential ($\Delta\Phi$) is defined as follows:

$$\Delta\Phi = \frac{j_{f,m}}{\sigma} + j_{f,m}R_{ohm,m}, \tag{3}$$

where m represents the index of the electrode particle−electrolyte interface, $j_{f,m}$ denotes the reaction current density at interface m, σ signifies the electrical conductivity of the electrode particles, and $R_{ohm,m}$ denotes the resistance excluding the electrode particle resistance. Notably, this resistance includes charge transfer impedance and contact resistance, which arises from insufficient contact caused by surface roughness, the presence of oxide layers, impurities, and other factors. The value of $R_{ohm,m}$ varies across interfaces.

Next, the reaction current density $\left(j_{f,m}\right)$ can be calculated using the Butler−Volmer equation:

$$j_{f,m} = i_{0,m}\left(exp\left(\frac{\alpha_a F}{RT}(\eta_m \pm \Delta\Phi)\right) - exp\left(-\frac{\alpha_c F}{RT}(\eta_m \pm \Delta\Phi)\right)\right), \tag{4}$$

where $i_{0,m}$ denotes the exchange current density at interface m, and η_m represents the overpotential at interface m. This overpotential can be calculated as follows:

$$i_0 = Fk_r C_e^{\alpha_a}\left(C_{s,max} - C_{s,surf}\right)^{\alpha_a} C_{s,surf}^{\alpha_c}, \tag{5}$$

$$\eta = \Phi_{s,surf} - \Phi_e - E(y), \tag{6}$$

where $\Phi_{s,surf}$ denotes the electrostatic potential of the electrode particle, and Φ_e represents the electrolyte potential at the electrode−electrolyte interface. Following this calculation, an optimization algorithm can be adopted to determine the $\Delta\Phi_{neg}$ and $\Delta\Phi_{pos}$ values, ensuring that the surface potential changes are consistent within the anode and within the cathode regions, while the sum of the local reaction current densities at the electrode–electrolyte interfaces in both regions matches the applied load current density:

$$\sum_{m=1}^{N_{pos}} j_{f_m,pos} = \sum_{m=1}^{N_{neg}} j_{f_m,neg} = I_{app}, \tag{7}$$

where N denotes the number of electrode particle−electrolyte interfaces involved in electrochemical reactions. Although the reaction current density $j_{f,m}$ varies across interfaces, the sum of individual densities equals the current density of constant current discharge.

The surface potential of electrode particles at the next time step is defined as follows:

$$\Phi_{s,surf,t+1} = \Phi_{s,surf,t} \pm \Delta\Phi_t. \tag{8}$$

This equation indicates that the potential change of electrode particles across different regions remains consistent for each time step. Furthermore, according to Equation (3), the magnitude of this potential change must satisfy the following condition:

$$|\Delta\Phi| > max\left\{\left|\frac{j_{f,m}}{\sigma_s}\right| \middle| m = 0, 1, 2, \ldots, N\right\}. \tag{9}$$

Hence, unlike conventional battery modeling methods [16,17], the potential change of electrode particles considered in this study is not directly determined by the reaction current and conductivity. Moreover, the discharge boundary condition affects all electrode particle−electrolyte interfaces simultaneously, rather than only the battery terminals near the current collectors. To address the difficulty encountered by the LBM framework in handling cases where electrode particles directly intersect with the boundaries at the terminals of the battery, the boundary conditions at the left and right terminals of the battery model are replaced by no-gradient boundary conditions, as detailed in Table 1.

Moreover, under the assumption that the electrical conductivities of all electrode particles are identical, factors such as particle size and shape influence the potential distribution. While heterogeneous electrode particle shapes lead to non-uniform potential propagation within the particles, differences in particle size cause variations in the total charge carried by each particle.

If these two factors—heterogeneous shapes and size variations—independently or alternately influence the potential change of electrode particles, the potentials at different locations within the battery will differ accordingly.

3. Implementation of the LBM-based 2D battery model

Previous studies [12,13] adopted the LBM framework to model ion and electron transport processes within a battery. The governing equations and their corresponding LBM formulations are summarized in

Table 1
Initial and boundary conditions selected for the electrochemical model of the lithium-ion battery.

Physical processes	Boundary conditions		
Electron transport in the electrolyte	$\Phi_e\vert_{x=L_n^-} = \Phi_e\vert_{x=L_n^+}$,	$\kappa_{e,neg}^{eff}\frac{\partial\Phi_e}{\partial x}\Big\vert_{x=L_n^-} = \kappa_{e,sep}^{eff}\frac{\partial\Phi_e}{\partial x}\Big\vert_{x=L_n^+}$,	$\frac{\partial\Phi_e}{\partial x}\Big\vert_{x=0} = \frac{\partial\Phi_e}{\partial x}\Big\vert_{x=L} = 0$,
	$\Phi_e\vert_{x=L_n+L_s^-} = \Phi_e\vert_{x=L_n+L_s^+}$,	$\kappa_{e,sep}^{eff}\frac{\partial\Phi_e}{\partial x}\Big\vert_{x=L_n+L_s^-} = \kappa_{e,pos}^{eff}\frac{\partial\Phi_e}{\partial x}\Big\vert_{x=L_n+L_s^+}$,	$\frac{\partial\Phi_e}{\partial y}\Big\vert_{y=0} = \frac{\partial\Phi_e}{\partial y}\Big\vert_{y=H} = 0$,
Lithium-ion transport in the electrolyte	$C_e\vert_{x=L_n^-} = C_e\vert_{x=L_n^+}$,	$D_{e,neg}^{eff}\frac{\partial C_e}{\partial x}\Big\vert_{x=L_n^-} = D_{e,sep}^{eff}\frac{\partial C_e}{\partial x}\Big\vert_{x=L_n^+}$,	$\frac{\partial C_e}{\partial x}\Big\vert_{x=0} = \frac{\partial C_e}{\partial x}\Big\vert_{x=L} = 0$,
	$C_e\vert_{x=L_n+L_s^-} = C_e\vert_{x=L_n+L_s^+}$,	$D_{e,sep}^{eff}\frac{\partial C_e}{\partial x}\Big\vert_{x=L_n+L_s^-} = D_{e,pos}^{eff}\frac{\partial C_e}{\partial x}\Big\vert_{x=L_n+L_s^+}$,	$\frac{\partial C_e}{\partial y}\Big\vert_{y=0} = \frac{\partial C_e}{\partial y}\Big\vert_{y=H} = 0$,

Table 2
Governing equations and corresponding lattice Boltzmann equations.

Governing equations	Lattice Boltzmann equations
$\frac{\partial C_s}{\partial t} - \nabla\cdot(D_s \nabla C_s) = 0,$	$h_{s,i}(\mathbf{x} + \mathbf{c}_i\Delta t, t + \Delta t) - h_{s,i}(\mathbf{x}, t) = -\frac{1}{\tau_h}\left[h_{s,i}(\mathbf{x}, t) - h_{s,i}^{eq}(\mathbf{x}, t)\right],$
$\nabla\cdot(\sigma\nabla\Phi_s) = 0,$	$g_{s,i}(\mathbf{x} + \mathbf{c}_i\Delta t, t + \Delta t) - g_{s,i}(\mathbf{x}, t) = -\frac{1}{\tau_D}\left[g_{s,i}(\mathbf{x}, t) - g_{s,i}^{eq}(\mathbf{x}, t)\right],$
$\frac{\partial C_e}{\partial t} - \nabla\times(D_e\nabla C_e) - \nabla\times\left(\frac{D_e zFC_e}{RT}\nabla\Phi_e\right) = 0,$	$h_{e,i}(\mathbf{x} + \mathbf{c}_i\Delta t, t + \Delta t) - h_{e,i}(\mathbf{x}, t) = -\frac{1}{\tau_\kappa}\left[h_{e,i}(\mathbf{x}, t) - h_{e,i}^{eq}(\mathbf{x}, t)\right] + \omega_i\frac{F\nabla\cdot(zD_e\nabla C_e)}{RT},$
$\nabla\cdot\kappa\nabla\phi_e + \nabla\cdot\left(\frac{D_e zFC_e}{RT}\nabla C_e\right) = 0,$	$g_{e,i}(\mathbf{x} + \mathbf{c}_i\Delta t, t + \Delta t) - g_{e,i}(\mathbf{x}, t) = -\frac{1}{\tau_D}\left[g_{e,i}(\mathbf{x}, t) - g_{e,i}^{eq}(\mathbf{x}, t)\right] + \frac{zF\mathbf{c}_i\cdot\nabla\Phi_e}{RT}g_{e,i}^{eq}(\mathbf{x}, t),$
$c_i = \{(0,0),(1,0),(0,-1),(-1,0),(0,1),\}, i = 0,1,2,3,4$	

Table 2.

3.1. LBM model

The LBM model employed in this study is based on the D2Q5 framework proposed in Reference 13 [13]. However, the equilibrium functions, relaxation times for concentration fields, and potential fields used in this study differ slightly from those presented in Reference 13 [13]. The relevant equations are simplified as follows:

$$h_i^{eq} = \omega_i\Phi, \tag{10}$$

$$g_i^{eq} = \omega_i C, \tag{11}$$

$$\tau_h = \frac{\sigma}{cs^2} + 0.5, \tag{12}$$

$$\tau_g = \frac{D}{cs^2} + 0.5, \tag{13}$$

where g denotes the concentration distribution function, h represents the potential distribution function, c signifies the lithium/lithium-ion concentration, and Φ denotes the electric potential. Further, ω_i denote weighting coefficients, where $\omega_0 = \frac{1}{3}$ and $\omega_{1-4} = \frac{1}{6}$. cs represents the lattice speed, with $cs^2 = \frac{1}{2}$. D denotes the diffusion coefficient for the concentration field, and σ indicates the conductivity of the potential field. Finally, τ_g and τ_h represent the relaxation times for the concentration and potential fields, respectively.

For the electrochemical reaction at the electrode particle−electrolyte interface, Section 2 has already delved into the potential change of electrode particles. Moving further, the changes in electrolyte potential and lithium/lithium-ion concentration are determined using the following equations:

$$\sigma\frac{\partial\Phi}{\partial n} = j_f, \tag{14}$$

$$D\frac{\partial C}{\partial n} = \frac{j_f}{zF}. \tag{15}$$

Fig. 3(b)presents a schematic of the reaction interface between the electrolyte phase and solid particles within the LBM framework. Here, the change in concentration or potential at the reaction interface is represented by the difference in the corresponding distribution functions perpendicular to the interface. For the electrolyte phase, based on Equations (14) and (15), the interface electrochemical reaction equation in the LBM framework is expressed as follows [12–14]:

$$h_{e,4} - h_{e,2} = j_f\left(1 + \frac{1}{2\tau_{\Phi_e}}\right), \tag{16}$$

$$g_{e,4} - g_{e,2} = \frac{j_f}{zF}\left(1 + \frac{1}{2\tau_{C_e}}\right), \tag{17}$$

In these equations, the distribution functions $h_{e,4}$ and $g_{e,4}$, representing the transfer from the electrode particle phase to the electrolyte phase, are initially unknown and can be solved using the following procedure. The non-equilibrium parts of the distribution functions exhibit opposite signs in opposite directions, implying that the sum of the distribution functions in opposite directions equals the sum of the corresponding equilibrium functions:

$$h_{e,4} + h_{e,2} = h_{e,4}^{eq} + h_{e,2}^{eq} = (\omega_2 + \omega_4)\Phi_e, \tag{18}$$

$$g_{e,4} + g_{e,2} = g_{e,4}^{eq} + g_{e,2}^{eq} = (\omega_2 + \omega_4)C_e. \tag{19}$$

By combining Equation (16) with Equation (18) and Equation (17) with Equation (19) and substituting the values of the weighting coefficients, the following relations can be derived:

$$\Phi_e = 6h_{e,2} + 3j_f\left(1 + \frac{1}{2\tau_{\Phi_e}}\right), \tag{20}$$

$$C_e = 6g_{e,2} + \frac{j_f}{zF}\left(1 + \frac{1}{2\tau_{C_e}}\right). \tag{21}$$

Given that $h_{e,2}$ and $g_{e,2}$ are known distribution functions governed by the initial conditions, Φ_e and C_e can be derived using Equations (20) and (21). These Φ_e and C_e values can then be substituted into Equations (18) and (19), respectively, to determine the unknown distribution functions $h_{e,4}$ and $g_{e,4}$. In this study, τ_Φ and τ_C in the above equations are replaced by the corresponding lithium-ion diffusion coefficient D and electrolyte conductivity σ to minimize modeling errors.

After completing these calculations, the concentration and potential at each time step are updated using the following equations:

$$C = \sum g_i, \tag{22}$$

$$\Phi = \sum h_i. \tag{23}$$

3.2. Medium node handling

Fig. 2 illustrates the structure of the battery model, with black regions representing the solid phase (electrode particles) and white regions denoting the liquid phase (electrolyte). To minimize the impact of electrode particle morphology variations on computational accuracy, electrode particles with uniform sizes and distributions are used. The total computational domain spans 1134 × 525 lattice units.

In our analysis, node handling at the electrode–electrolyte interface is implemented using the method proposed in Reference 21 [21]. As illustrated in Fig. 3(a), nodes at the interface are classified into 12 types: flat wall nodes (types 1–4), located along linear boundaries; convex nodes (types 5–8), formed at the intersection of two linear boundaries;

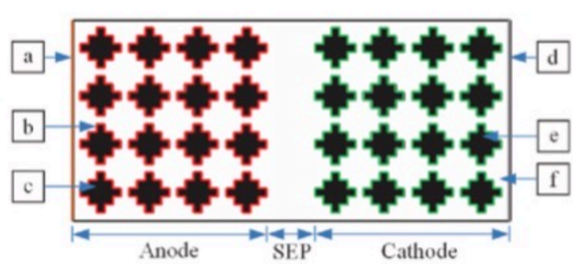

Fig. 2. Two-Dimensional Battery Model with uniform particle diameters. (a) Anode current collector, (b) reactive interface, (c) anode active material particles, (d) cathode current collector, (e) cathode active material particles, and (f) electrolyte.

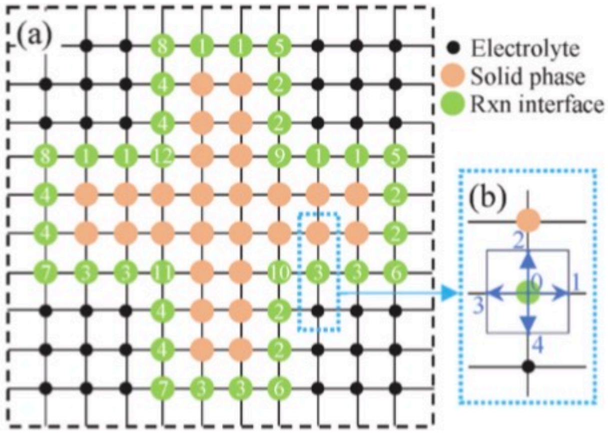

Fig. 3. Design of the porous geometry. (a) Illustration of the 12 types of boundary nodes and (b) nodes at the reactive interface.

and concave nodes (types 9–12).

3.3. Optimization algorithm for surface potential determination

To determine the uniform surface potential change $\Delta\Phi$ of electrode particles in the anode and cathode regions at each time step, we adopt Brent's method, a well-established optimization algorithm for one-dimensional problems. The objective function is formulated as:

$$f(\Delta\Phi) = \left(\sum_{m-1}^{N} j_{f,m}(\Delta\Phi) - I_{app} \right)^2 \tag{24}$$

where $j_{f,m}$ local reaction current density at interface m, dependent on the overpotential which includes $\Delta\Phi$. The optimization seeks the $\Delta\Phi$ value that minimizes the discrepancy between the sum of local reaction currents and the externally applied discharge current density I_{app}, thereby enforcing current conservation.

3.4. Material parameters

The physical parameters employed in the modeling process, along with their corresponding dimensionless lattice values, are outlined in Table 3.

The physical properties of the electrolyte remain consistent across all regions of the battery model, with a diffusion coefficient of $D_e = 1.5 \times 10^{-9}$ m^2s^{-1}, conductivity of $\sigma = 11.47$ Sm^{-1}, and initial lithium concentration of $C_e^0 = 2000$ $molm^{-3}$. Additionally, the length of the separator region is 25 μm, while the temperature is 298 K. The height of the model is $H = 25$ μm. Each time step corresponds to 0.0013 s, and the discharge current density is $I_{app} = 9$ Am^{-2}.

Table 3
Physical parameters and corresponding dimensionless lattice units used in the simulation.

Symbol (Unit)	Negative	Positive
α	0.5	0.5
ε_s	0.3545	0.3545
ε_e	0.6	0.6
D_s (m^2s^{-1})	7.8×10^{-13} (2.6636×10^{-2})	2×10^{-12} (6.8299×10^{-2})
C_s^{max} $(molm^{-3})$	26,390 (9.4048×10^{-5})	22,860 (8.1468×10^{-5})
C_s^0 $(molm^{-3})$	14,030 (5×10^{-5})	3900 (1.3898×10^{-5})
C_e^0 $(molm^{-3})$	2000 (7.1275×10^{-6})	2000 (7.1275×10^{-6})
σ (Sm^{-1})	100 (2.4549)	3.8 (9.3285×10^{-2})
κ_r $\left(mol^{-0.5}m^{2.5}s^{-1}\right)$	4.4×10^{-11} (4.9941×10^{-3})	4.8×10^{-11} (5.4481×10^{-3})
L (μm)	100 (1.9841×10^{-7})	100 (1.9841×10^{-7})
r (μm)	8.3 (1.6468×10^{-6})	8.3 (1.6468×10^{-6})

The equilibrium function for the positive electrode particles is as follows:

$$E_n(V) = -0.8875y_1^5 + 3.1733y_1^4 - 5.4406y_1^3 + 5.8375y_1^2 - 3.9501y_1 + 1.1597, \tag{25}$$

$$E_p(V) = \begin{cases} -4.325y_2 + 5.033175, 0.1 \le y_2 < 0.195 \\ -247.9y_2^3 + 186.7y_2^2 - 47.2y_2 + 8.13268, 0.195 \le y_2 < 0.255 \\ -0.08218598958y_2 + 4.14728206484, 0.255 \le y_2 < 0.495 \\ -179.2y_2^3 + 255.8y_2^2 - 121.5y_2 + 23.3064026, 0.495 \le y_2 < 0.555 \\ 375y_2^3 - 794.4y_2^2 + 531.9y_2 - 110.5853799, 0.555 \le y_2 < 0.715 \end{cases}, \tag{26}$$

where y_1 and y_2 represent the ratios of the surface concentrations of the anode and cathode particles to their respective maximum concentrations. Specifically, they are defined as $y_1 = \frac{C_{s,n,surf}}{C_{s,n}^{max}}$ and $y_2 = \frac{C_{s,p,surf}}{C_{s,p}^{max}}$.

All electrical parameters in the model are non-dimensionalized using a reference voltage $U_{ref} = 1V$. Based on the SI definition of voltage, we derived the corresponding scaling factors for other electrical parameters (such as current and current density) by linking them with characteristic scales for length, time, mass, and substance amount, thus ensuring dimensional consistency.

4. Simulation and results

4.1. Model validation

To validate the effectiveness and accuracy of the proposed model, its results were compared with those derived from the model proposed in Reference 13[13] and those obtained using the traditional finite element method (FEM). For the FEM analysis of discharge performance, a 1D isothermal model based on the P2D mechanism was developed using the commercial software COMSOL-Multiphysics. The initial conditions and physical parameters used in the LBM study were applied in this case as well.

In the current LBM framework, the discharge voltage is defined as the difference between the average potential of the anode particles and that of the cathode particles:

$$U = \overline{\Phi}_{s,p,surf} - \overline{\Phi}_{s,n,surf}. \tag{27}$$

Notably, this potential is not a time-series potential but represents the change in potential corresponding to the depth of discharge (DOD). Because the state of charge (SOC) differs between the anode and cathode particles, interpolation based on the relationship between potential and DOD is required before using it in Equation (25) for further calculations. The SOC and DOD are defined as follows:

$$SOC = \frac{\overline{C_s} - C_{s,min}}{Cs, min_{s,max}} \tag{28}$$

$$DOD = 1 - SOC \tag{29}$$

The $C_{s,min}$ and $C_{s,max}$ represent the average embedded lithium concentrations at 0 % and 100 % SOC, respectively.

As part of our analysis, the voltage responses of the models at different discharge rates (1C, 5C, and 10C) were computed, and the results are illustrated in Fig. 4(a). Owing to the longer duration of the 1C discharge, more data points were generated. Hence, for clarity, data under the 1C discharge condition were recorded once every five iterations, while those under the 5C and 10C discharge conditions were recorded at each iteration.

Given that the model parameters utilized in Ref. [13] differ from those adopted in this study, the voltage response curve derived from the

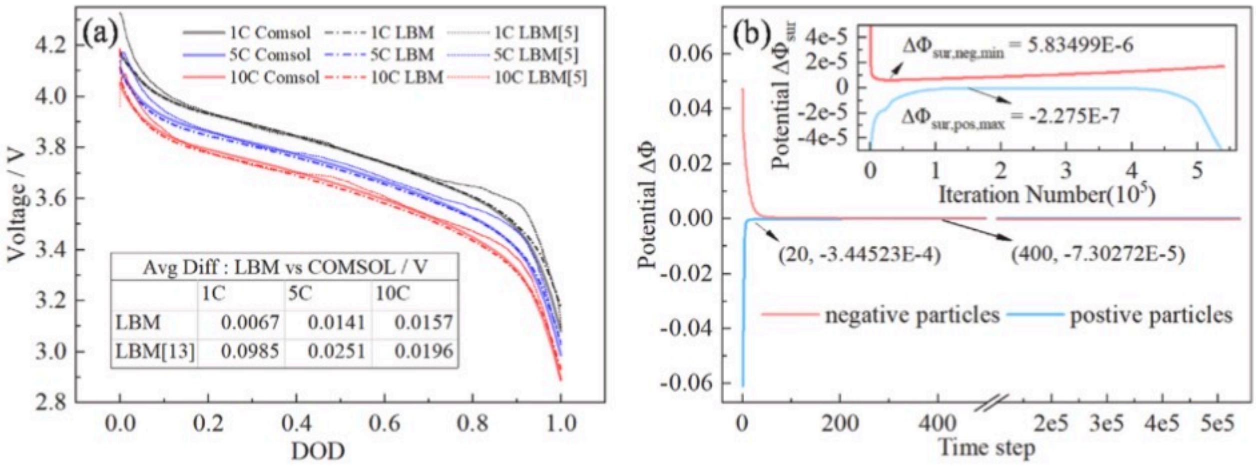

Fig. 4. Changes in voltage/potential during the discharge process. (a) Voltage responses at different discharge rates. (b) Lattice values of surface potential changes at each time step under 5C discharge rate.

model proposed in Reference 13 was recomputed based on the parameters used in this study [13]:

$$result_{LBM[13]} = \left(\frac{result^{old}_{LBM[13]} - result_{COMSOL[13]}}{result_{COMSOL[13]}} \right) * result_{COMSOL}, \tag{30}$$

where $result^{old}_{LBM[13]}$ and $result_{COMSOL[13]}$ denote the voltage response curves obtained from the model previously proposed in Ref. [13] using the COMSOL software and LBM, respectively. Further, $result_{LBM[13]}$ and $result_{COMSOL}$ represent the voltage response curves obtained from the model proposed in Ref. [13] using the LBM framework and COMSOL software, respectively, based on the parameters utilized in this study.

Fig. 4(a) indicates that the voltage response of the 2D battery model proposed in this study generally aligns with the COMSOL simulation results, exhibiting the highest accuracy at the 1C discharge rate. The accuracy of the proposed model also considerably surpasses that of the model previously proposed in Ref. [13], particularly at the beginning and near the end of the discharge process [13].

Furthermore, the error of the proposed model increases with increasing discharge rates, whereas that of the model proposed in Reference 13 initially increases and then decreases with rising discharge rates [13].Specifically, as the discharge increases, the electrochemical and physical processes occurring within the battery become more complex and dynamic, leading to a typical decline in model accuracy. However, the proposed model demonstrates good numerical stability even at high discharge rates.

In terms of the voltage response trend at different discharge rates, the proposed model exhibits a slower response compared to the COMSOL model. Specifically, when the voltage trend begins to change, the voltage variation in the COMSOL model is more pronounced than that in the proposed model. The model proposed in Reference 13 exhibits a similar trend [13]; however, the magnitude of the voltage variation in this case initially decreases and then increases. Once the voltage trend stabilizes, the voltage response of the model proposed in Reference 13 aligns with that of the COMSOL model [13]; however, the voltage response of the proposed model exhibits a relatively small and stable error compared to the COMSOL model. This trend is particularly evident in the voltage responses of the proposed model under the 5C and 10C discharge conditions during the mid and later sections of the voltage curve.

Additionally, the initial voltage values predicted by the proposed model remain consistent across the different discharge rates, whereas those of the COMSOL model exhibit noticeable differences. This implies that, in the early discharge phase, the voltage response of the proposed model displays a rapid drop, where the voltage quickly declines to a value close to the initial voltage of the COMSOL model (within approximately 1000 time steps). After this initial drop, the voltage trend of the proposed model aligns closely with that of the COMSOL model. This phenomenon may be attributed to the omission of certain computational results in the COMSOL model.

The change in the surface potentials ($\Delta\Phi$) of the anode and cathode particles at each time step during the discharge cycle, $\Delta\Phi$, is calculated using an optimization algorithm based on the discharge current density. The results reveal that the potential change is uniform across all electrode particles within the same region at any given time. As illustrated in Fig. 4(b), the trend exhibited by $\Delta\Phi$ can be divided into two distinct phases.

Initially, during time steps 0 to 20, $\Delta\Phi$ rapidly approaches zero owing to disturbances caused by the instability of the LBM system, which is unrelated to the actual battery behavior. Following this phase, $\Delta\Phi$ stabilizes and fluctuates within a narrow range, following a trend similar to the equilibrium potential of the electrode.

The average surface current density of the electrode particles under 1C discharge rate is 1.053596 A/m^2, with a corresponding lattice value of 5.131814×10^{-9}. Using the conductivity values detailed in Table 3, the calculated $j_{f,m}/\sigma_s$ lattice values for the anode and cathode regions under 5C discharge rate are 1.0452×10^{-8} and 2.7506×10^{-7}, respectively. As depicted in Fig. 4(b), the absolute maximum values of $\Delta\Phi$ for the anode and cathode regions are 5.83499×10^{-6} and 2.275×10^{-7}, evidently, the anode region satisfies the requirements of Equation (9).

For the cathode region, over the time steps between 20 and 540,000, $\Delta\Phi$ values fall below 2.7506×10^{-7} for 193,341 steps, representing 35.8 % of the total. However, in terms of the mean value, the absolute average of $\Delta\Phi$ over the time steps from 20 to 540,000 is 5.3664×10^{-6}, which likewise meets the conditions of Equation (9).

4.2. Voltage and current changes during constant resistance discharge

Based on the modeling approach adopted in this study, the influence of an external circuit's load on battery performance during discharge can be represented by the change in the surface potential of the negative electrode particles, denoted as $\Delta\Phi_{neg}$. Once this value is determined, the total current density in the negative electrode region can be computed. Following this, the change in the surface potential of positive electrode particles, denoted as $\Delta\Phi_{pos}$, can be derived using an optimization algorithm. This approach facilitates a qualitative analysis of the voltage and current changes occurring during constant resistance discharge.

Let us assume a load circuit with resistance R. This circuit results in a change of $\Delta\Phi_{neg} = 6.15924 \times 10^{-6}$ in the surface potential of the negative electrode particles at each time step. Conversely, another load circuit with a resistance of $R/2$ or $R/3$ results in a surface potential change of $\Delta\Phi_{neg} = 1.2318 \times 10^{-5}$ V or $\Delta\Phi_{neg} = 1.8478 \times 10^{-5}$ V, respectively, at each time step. Fig. 5 represent the changes in battery voltage and current density during constant resistance discharge.

As depicted in Fig. 5(a), higher external circuit resistance results in a slower discharge rate and, consequently, a longer discharge time. For a given discharge time, a higher external circuit resistance leads to a lower load voltage. However, the general trend of the voltage response curves for different loads remains similar, which is governed by the equilibrium potential of the electrode materials.

Fig. 5(b) plots the total current density changes in both the negative and positive electrode regions for different loads. Throughout the discharge cycle, the current density exhibits a monotonically decreasing trend. Initially, the current density decreases rapidly, followed by a significant deceleration in the rate of decline, approaching a near-constant discharge. Subsequently, the current density declines again at a relatively faster rate until nearing the end of discharge, where the rate of decline slows. The model simulation results align well with typical constant-resistance discharge behavior.

4.3. Analysis of the discharge process in battery models with variations in electrode particle diameters

The following discharge process analysis is based on the 5C discharge data. In Fig. 6(a), the black regions represent the solid phase (electrode particles), while the white regions represent the liquid phase (electrolyte). To differentiate these results from those in Ref. [19], the electrode particles are modeled with two distinct particle sizes [19]. Each particle size category consists of an equal number of uniformly sized particles.

The temporal evolution of lithium/lithium-ion concentration within the lithium battery electrode is illustrated in Fig. 6(b–f). The lithium concentration is normalized by the saturated lithium concentration of the negative electrode particles, which is 26,390 mol/m^3. Fig. 6(c) and (e) illustrate the lithium concentration distribution at two different time steps, ranging from 3000 to 580,000. Notably, the conductivity of the electrode particles is high, and the potential gradient within homogeneous electrode particles is not significant. At the beginning of the discharge, lithium is primarily concentrated within the negative electrode particles. By the end of the discharge, lithium deintercalation occurs from the negative electrode particles, while lithium accumulation is observed in the positive electrode particles.

Fig. 6(b), (d), and 6(f) depict the lithium-ion concentration distribution at three different time steps, ranging from 300 to 580,000. At 300 time steps, lithium ions from the negative electrode region diffuse uniformly outward from each electrode particle, with the concentration gradually decreasing from high to low, centered around each particle. In the positive electrode region, lithium ions are uniformly added to the surfaces of electrode particles, and the concentration increases from low to high, with the electrode particles at the center. This behavior occurs because, in this study, the discharge boundary conditions are applied simultaneously to the surfaces of all electrode particles.

At 3000 time steps, the lithium-ion concentration distribution reflects the diffusion and migration processes of lithium ions from the negative electrode to the positive electrode. In the negative electrode region, the concentration is higher at the center of each particle and lower at the periphery, while in the positive electrode region, the concentration is lower at the center of each particle and higher at the periphery. Furthermore, a trend of lithium-ion movement from the negative to the positive electrode is apparent. Along the horizontal direction, the lithium-ion concentration in the central region of the negative electrode is lower compared to that near the left edge.

At 580,000 time steps, the lithium-ion concentration distribution transitions from a radial pattern to a stepped distribution, with the concentration gradually decreasing from left to right. This shift results from the migration and diffusion of lithium ions from the negative to the positive electrode region, driven by factors such as electrolyte potential and concentration gradients.

Fig. 7(a) illustrates the surface potential changes of anode and cathode particles during the 5C discharge cycle. Evidently, although the potential change of the particles remains identical at each time step in the anode and, separately, in the cathode regions, the surface potentials at different locations within the same region vary.

The average surface potential of the anode and cathode regions is close to its minimum value, indicating that most electrode particles possess surface potentials similar to and close to this minimum. This minimum value primarily occurs on the surfaces of larger electrode particles, suggesting that the equipotential assumption for electrode particles, as proposed in Ref. [19], is largely valid when particle sizes are uniform [19].

However, a few interfaces exhibit surface potentials that deviate substantially from the average. These interfaces correspond to the maximum surface potential in the anode and cathode regions. The observed deviations are primarily found on the surfaces of smaller electrode particles, which is attributed to the differences in electrochemical reaction rates on particles of varying sizes. This suggests that the equipotential assumption becomes invalid when substantial size differences exist between electrode particles [19].

The changes in the surface potentials of the anode and cathode regions resemble the changes in the equilibrium functions of the corresponding electrode particles.

Fig. 7(b) depicts the variation in reaction current density at the electrode particle–electrolyte interface. As illustrated, the absolute average value of the reaction current density for both the anode and cathode regions remains consistent and constant throughout the 5C discharge cycle. This indicates that the system maintains both charge

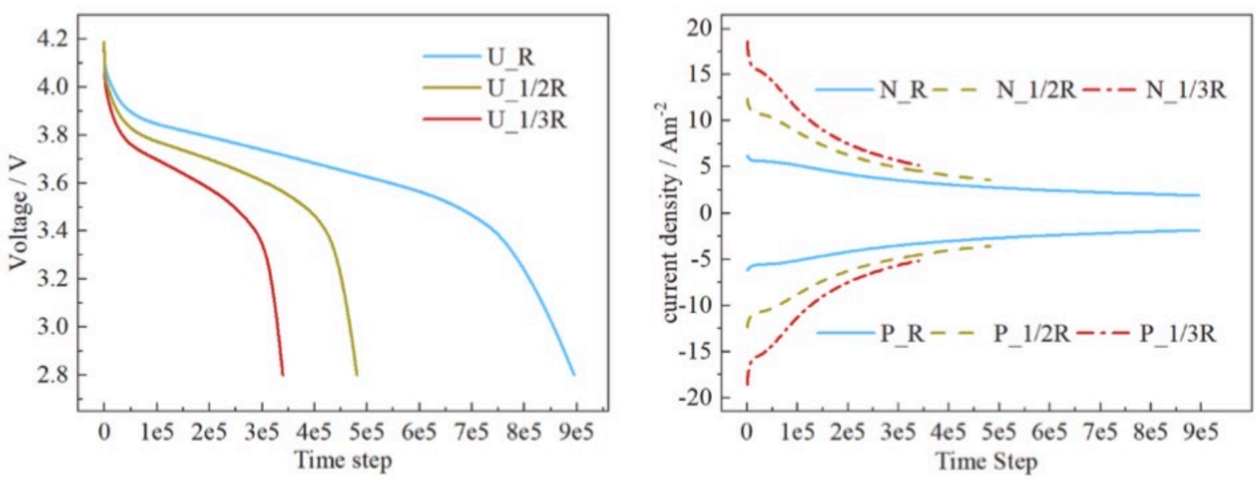

Fig. 5. Voltage and current changes during constant resistance discharge. (a) Voltage response and (b) current density changes.

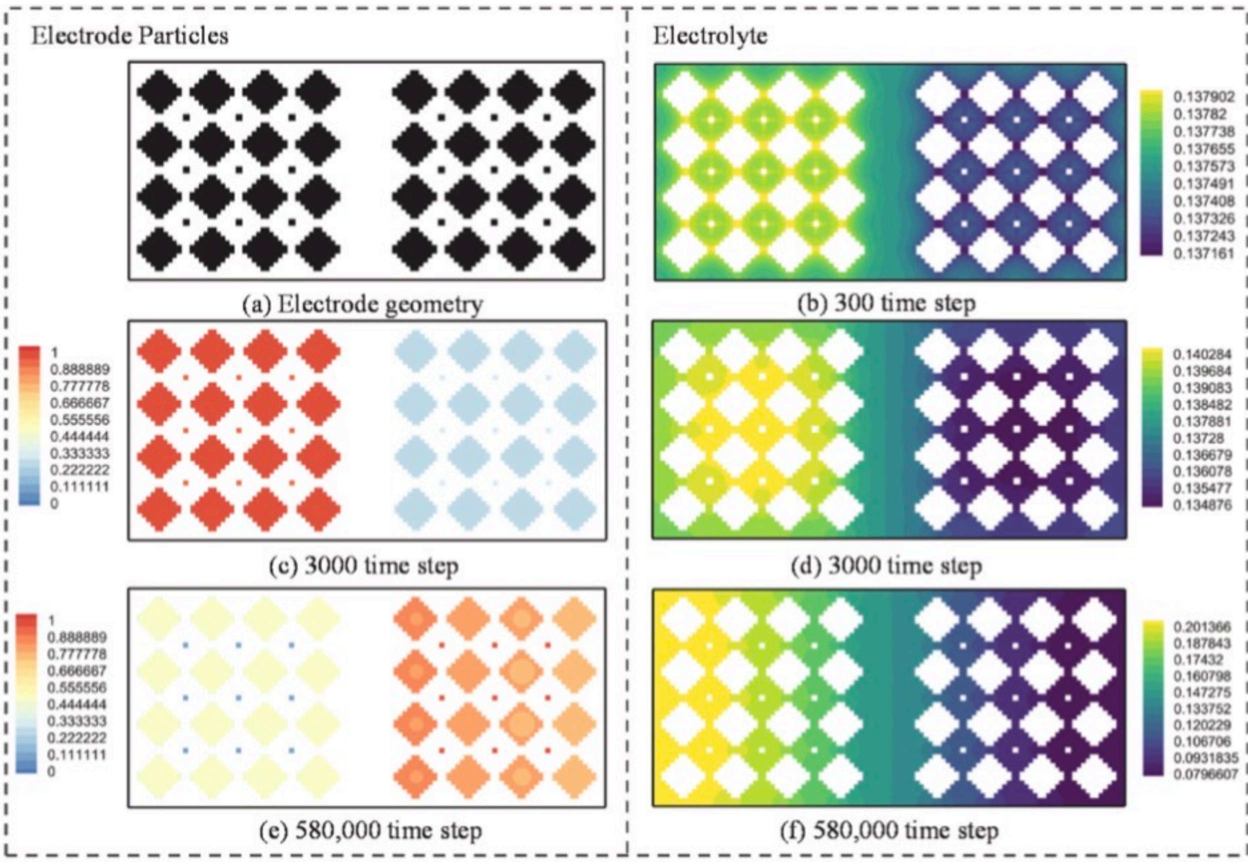

Fig. 6. Visualization of lithium concentration evolution during discharge. (a) Electrode geometry with non-uniform particle sizes, and (b)–(f) normalized lithium concentration distribution in the solid and electrolyte phases at different time steps.

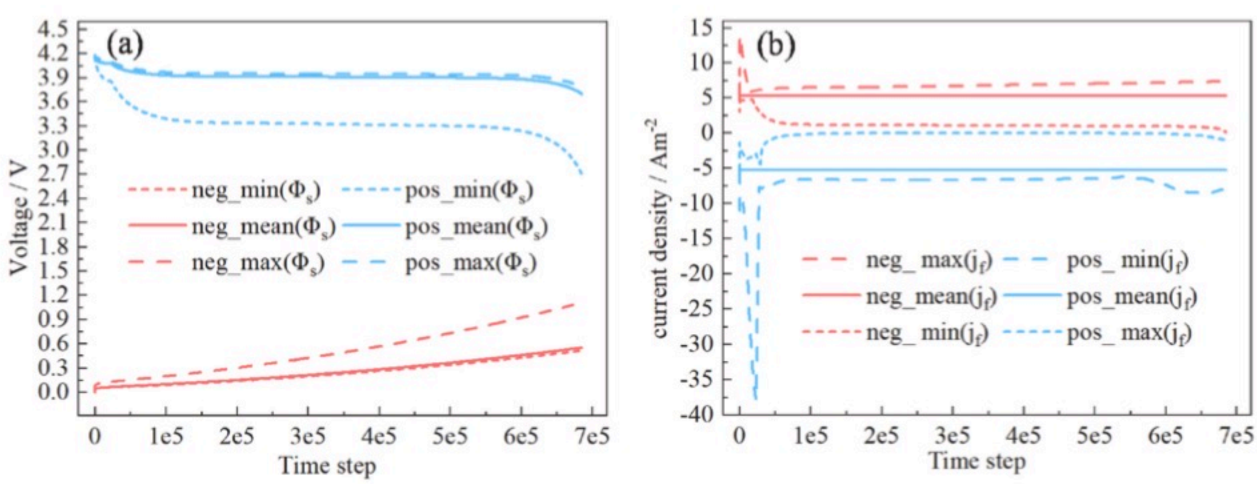

Fig. 7. Potential and local reaction current density changes during the 5C discharge process. (a) Changes in the surface potentials of negative and positive electrode particles. (b) Changes in the maximum, minimum, and average values of the local current density in the negative/positive electrode regions at each time step.

conservation and current conservation.

The evolution of the extreme values of reaction current density, as depicted in Fig. 7(b), indicates that under the initial conditions, the absolute value of the local reaction current is identical across all electrode particle–electrolyte interface positions. However, as the discharge progresses, differences in these extreme values begin to appear, and the evolution process can be divided into three distinct phases:

Rapid Fluctuation Phase: The absolute value of the local reaction current density quickly deviates from the regional average but subsequently returns toward the mean.

Stable Phase: The local reaction current density largely stabilizes, exhibiting only gradual, minor variations.

Closing Phase: The trend in local reaction current density differs between the anode and cathode regions. In the anode region, the reaction current density deviates from the mean, while in the cathode region, it converges toward the mean.

The duration and magnitude of these phases vary depending on the position of each particle, an aspect that will be analyzed further (Fig. 8).

Next, the center positions of large-sized electrode particles, denoted as N_i (i = 1, 2, 3, and 4), and the center positions of small-sized electrode particles, denoted as P_i (i = 1, 2, 3, and 4), are marked in Fig. 8(b) and (c). Using these positions as reference points, the changes in lithium concentration at various locations during discharge are analyzed, as illustrated in Fig. 8. In Fig. 8(a), the red lines represent lithium

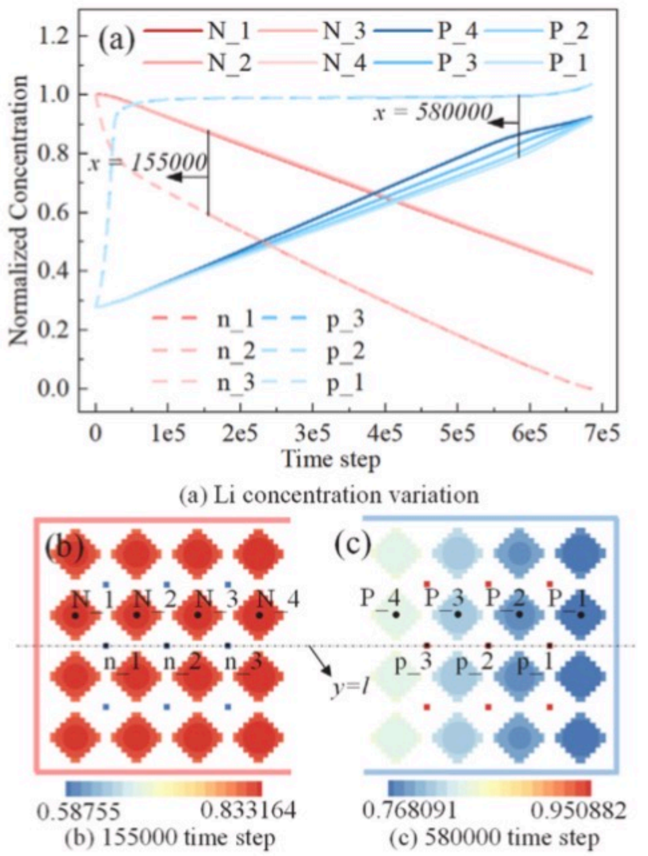

Fig. 8. Variations in lithium concentration at different time steps. (a) Changes in lithium concentration with different particle positions. (b) Lithium concentration distribution at 155,000 time steps in the negative electrode region. (c) Lithium concentration distribution at 580,000 time steps in the positive electrode region.

concentration changes in the negative electrode particles, while the blue lines denote the lithium concentration changes in the positive electrode particles. Within the same electrode region, particles of the same size exhibit differences in lithium concentration changes. Specifically, in the positive electrode region, large particles experience more pronounced variations in concentration, primarily influenced by their position.

In the negative electrode region, as illustrated in Figs. 7(b) and 8(a), higher reaction current densities initially appear on the surfaces of smaller particles at the onset of the discharge, and lithium is rapidly deintercalated from these small particles. As discharge continues, the high reaction current density decreases, and the lithium deintercalation rate in small particles gradually slows, approaching that in the larger particles. At this stage, higher reaction current densities shift to the surfaces of larger particles. Toward the end of the discharge process, almost all of the lithium present within the small particles is completely deintercalated, while some lithium remains within the larger particles. Throughout the discharge process, the lower reaction current densities also remain on the surfaces of large particles, maintaining relatively stable values.

In the positive electrode region, as depicted in Figs. 7(b) and 8(a), higher absolute reaction current densities first appear on the surfaces of smaller particles, where lithium rapidly intercalates, causing a steep decline in their potential. Once lithium is almost fully intercalated into the smaller particles, the higher current density shifts to the surfaces of larger particles, while the surface current on the smaller particles becomes negligible. Toward the end of the discharge process, fluctuations in the maximum reaction current occur in the positive electrode region. The lithium intercalation rate of large particles varies depending on their location. Specifically, particles near the boundaries intercalate lithium more rapidly than those near the separator. At the end of the discharge phase, the lithium concentrations in large particles at different locations tend to converge.

Fig. 8(b) and (c) present the lithium concentration contour plots for the negative electrode region at 155,000 time steps and the positive electrode region at 580,000 time steps, respectively. Evidently, as the discharge progresses, lithium is depleted from the surfaces of the negative electrode particles and accumulates on the surfaces of the positive electrode particles. During diffusion within individual particles, a pronounced lithium concentration gradient develops in the larger particles, while the lithium concentration remains more uniformly distributed in the smaller particles.

Along the vertical direction, the lithium concentration distribution within the larger particles at different positions exhibits mirror symmetry along the central axis $y = l$. In particular, in the cathode region, particles near the central axis display higher lithium concentrations compared to those near the top and bottom boundaries, while the opposite trend is observed in the anode region. This pattern arises owing to the zero-gradient boundary conditions applied at the top and bottom edges of the model.

Along the horizontal direction, significant differences in lithium concentration are observed across larger particles at different positions. In the anode region, the lithium concentration decreases sequentially from right to left. Conversely, in the cathode region, this concentration decreases sequentially from left to right. These differences result from the movement of lithium ions through the electrolyte from the negative electrode region to the positive electrode region, leading to variations in electrochemical reaction rates on the surfaces of particles based on their positions. Specifically, the core electrochemical reaction zone in the cathode shifts toward the separator, while in the anode, it moves toward the current collector. This phenomenon aligns with the conclusions of Reference [22] but shows partial differences from the results in Ref. [19]. This discrepancy may arise from the use of boundary conditions in Ref. [19] similar to those applied near the current collector boundary in Ref. [13].

5. Conclusion

The model developed in this study was validated by comparing its voltage responses at different discharge rates (1C, 5C, and 10C) with those of an FEM-based model. Notably, the simulation results of the model demonstrated close alignment with those obtained using COMSOL under the same initial conditions and parameter settings. In particular, at the 1C discharge rate, the model error was only 0.0067 V, better than that reported in Reference 13 [13]. However, when the voltage response trend varied, the voltage variation predicted by our model was slower and smaller than that predicted by the COMSOL model, likely owing to the limitations inherent in the 2D model.

The change in the surface potentials ($\Delta\Phi$) of the anode and cathode particles at each time step during the discharge cycle, $\Delta\Phi$, is calculated using an optimization algorithm based on the discharge current density. The results reveal that the potential change is uniform across all electrode particles within the same region at any given time. $\Delta\Phi$ following a trend similar to the equilibrium potential of the electrode. Moreover, its absolute value exceeds the potential derived from conductivity and the absolute value of reaction current density.

Based on the modeling approach adopted in this study, the influence of an external circuit's load on battery performance during discharge can be represented by the change in the surface potential of the negative electrode particles. A qualitative analysis of the voltage and current changes occurring during constant resistance discharge shows that the model simulation results align well with typical constant-resistance discharge behavior.

An analysis of the discharge process in battery models with variations in electrode particle diameters was subsequently conducted, with a focus on the resulting changes in local lithium/lithium-ion concentration and potential during discharge.

At each time step, all electrode particles within the same region undergo the same surface potential change simultaneously. However, the surface potentials at different positions on the electrode particles vary, influenced by factors such as potential transmission, position and particle size. When electrode particles are uniform in size, the surface potentials of different electrode particles are essentially the same, validating the equipotential assumption proposed in Ref. [19]. When electrode particles vary significantly in size, smaller particles in the positive electrode region exhibit rapid lithium intercalation, while smaller particles in the negative electrode region undergo rapid lithium deintercalation, both resulting in large potential changes on these respective small particles.

The predicted variation in local lithium concentration indicates that, along the horizontal direction, the movement of lithium ions through the electrolyte from the negative electrode region to the positive electrode region causes the core electrochemical reaction zone in the cathode to shift toward the separator, while in the anode, it moves toward the current collector. This phenomenon aligns with the conclusions of Reference [22] but shows partial differences from the results in Ref. [19]. This discrepancy may arise from the use of boundary conditions in Ref. [19] similar to those applied near the current collector boundary in Ref. [13].

Meanwhile, the local variation in lithium-ion concentration evolves from multi-center radial uniform diffusion to gradient diffusion, with the lithium-ion concentration decreasing from the negative electrode region toward the positive electrode region.

CRediT authorship contribution statement

Ming-Dai Yang: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Sheng-Qiao Hu:** Visualization, Software, Formal analysis, Conceptualization. **Chang-Ping Li:** Visualization, Resources. **Yein Kwak:** Supervision, Resources, Project administration. **Tae Jo Ko:** Supervision, Resources, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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Nomenclature

A: Electrode plate area, m^2
C: Lithium/Lithium-ion concentration, $molm^{-3}$
cs: Lattice sound speed
c_i: Discretized velocity direction velocity direction
D: Lithium/Lithium-ion diffusivity, m^2s^{-1}
d: Particle radius, μm
E: Electrode equilibrium potential, V
F: Faraday's constant, 96487 $Cmol^{-1}$
g: Concentration distribution function
H: Height of the calculation domain, m
h: Potential distribution function
I_{app}: Load current density, Am^{-2}
i_0: Exchange current density, Am^{-2}
j: Reaction flux, $molm^{-2}s^{-1}$
j_f: Local volumetric current density, Am^{-3}
k: Reaction rate constant, $mol^{-0.5}m^{2.5}s^{-1}$
L: Length of the calculation domain, m
m: Index or integer, dimensionless

N: Total number of cathode/anode particles
n: Normal vector
R: Universal gas constant, 8.31451 $Jmol^{-1}\kappa^{-1}$
r: Radial coordinate, m
T: Temperature, K
t: Time, s
Δt: Time step
U: Terminal voltage of battery cell, V
Δx: Space step
x: lD coordinate across the cell, m
z: lD coordinate across electrode/separator, m
ε: Volume fraction
α_a, α_c: Charge transfer coefficient, 0.5, 0.5
σ: Conductivity, Sm^{-1}
τ: Relaxation time
Φ: Potential, V
η: Activation overpotential, V

Superscripts

eq: Equilibrium
neq: Non-equilibrium
max: Maximum
min: Minimum

Subscripts

e: Electrolyte phase
i: Direction of the lattice velocity
l: Lattice unit
Li: Lithium ion
p: Physical unit
surf: Surface
s: Active material phase
neg: Negative area
sep: Separator area
pos: Positive area